

Gebze Technical University

Dep. of Mechanical Engineering

Statics (ME 231)

Lecture 12:

Cables

By Saeed Lotfan

Cables

Flexible cables and chains combine strength with lightness and often are used in structures for **support** and to **transmit** loads from one member to another.

- In support applications, cables form the main load-carrying element of the structure. For these cases the weight of the cable can be neglected because the weight is often small compared to the load the cable will support.
- In transmission applications, the cable weight may become important and must be included in the analysis.

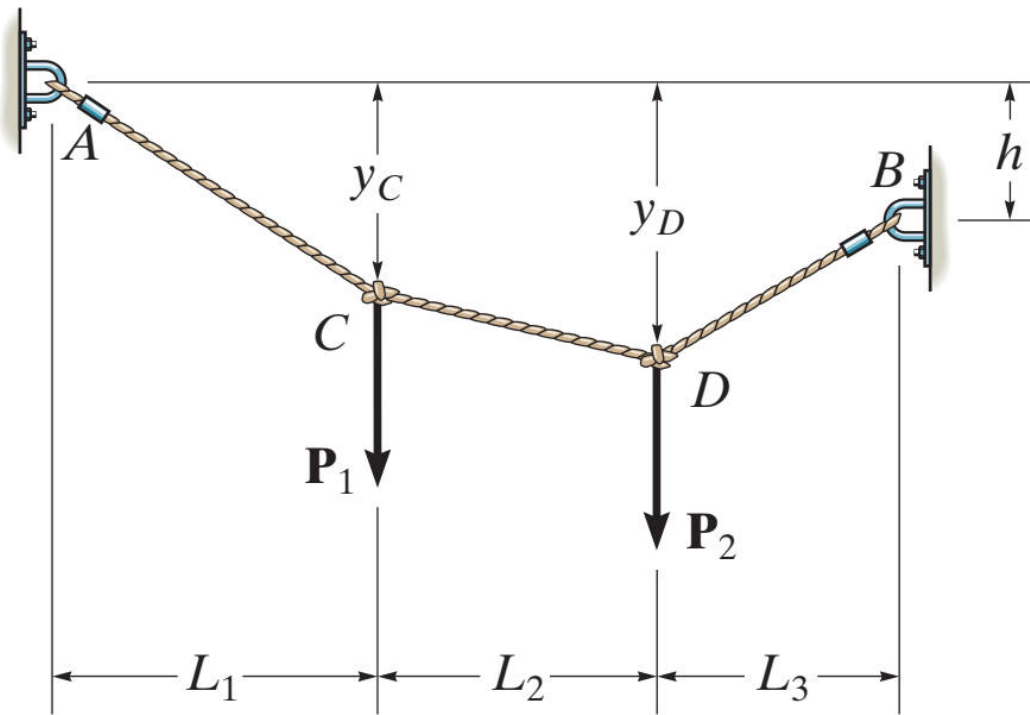
Cables

Assumptions in modeling:

The cable is *perfectly flexible* and *inextensible*. Due to its flexibility, the cable offers no resistance to bending, and therefore, the tensile force acting in the cable is always tangent to the cable at points along its length. Being inextensible, the cable has a constant length both before and after the load is applied. As a result, once the load is applied, the geometry of the cable remains unchanged, and the cable or a segment of it can be treated as a rigid body.

Cables Subjected to Concentrated Loads

When a cable of negligible weight supports several concentrated loads, the cable takes the form of several straight-line segments, each of which is subjected to a constant tensile force.



Example: Determine the tension in each segment of the cable shown below.

$$\sum F_x = 0 \quad A_x = -E_x$$

$$\sum F_y = 0 \quad A_y + E_y = 4 + 15 + 3 \text{ kN}$$

$$\sum M_A = 0 \quad -4(3) - 15(8) - 3(16) + E_y(18) = 0$$

$$E_y \checkmark, A_y \checkmark$$

point A:

$$\sum F_x = 0 \quad A_x + T_{AB} \cos \theta_1 = 0$$

$$\sum F_y = 0 \quad A_y - T_{AB} \sin \theta_1 = 0$$

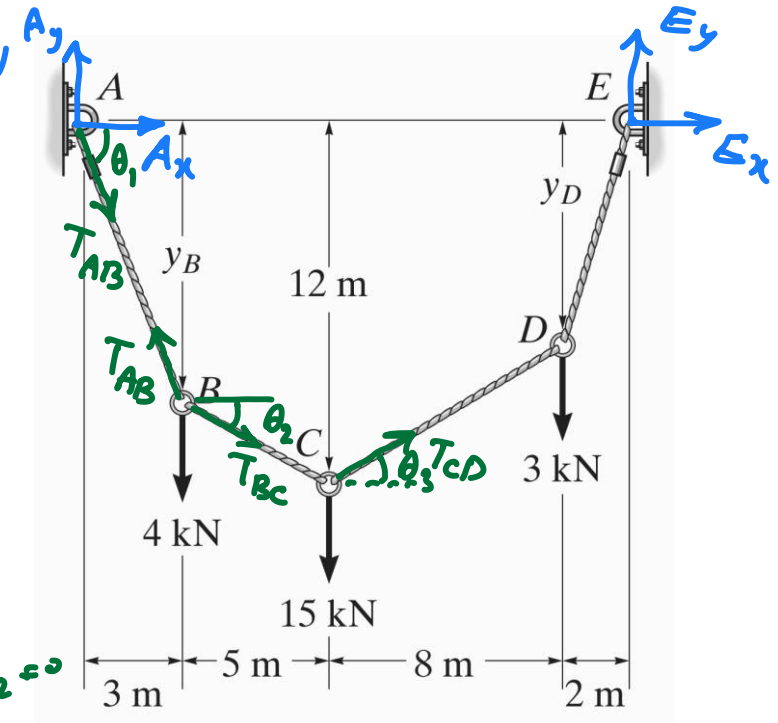
point B:

$$\sum F_x = 0 \quad T_{BC} \cos \theta_2 = T_{AB} \sin \theta_1$$

$$\sum F_y = 0 \quad T_{AB} \cos \theta_1 - 4 - T_{BC} \sin \theta_2 = 0$$

point C: ; point D:

$$\sum F_x = 0 \quad \dots$$

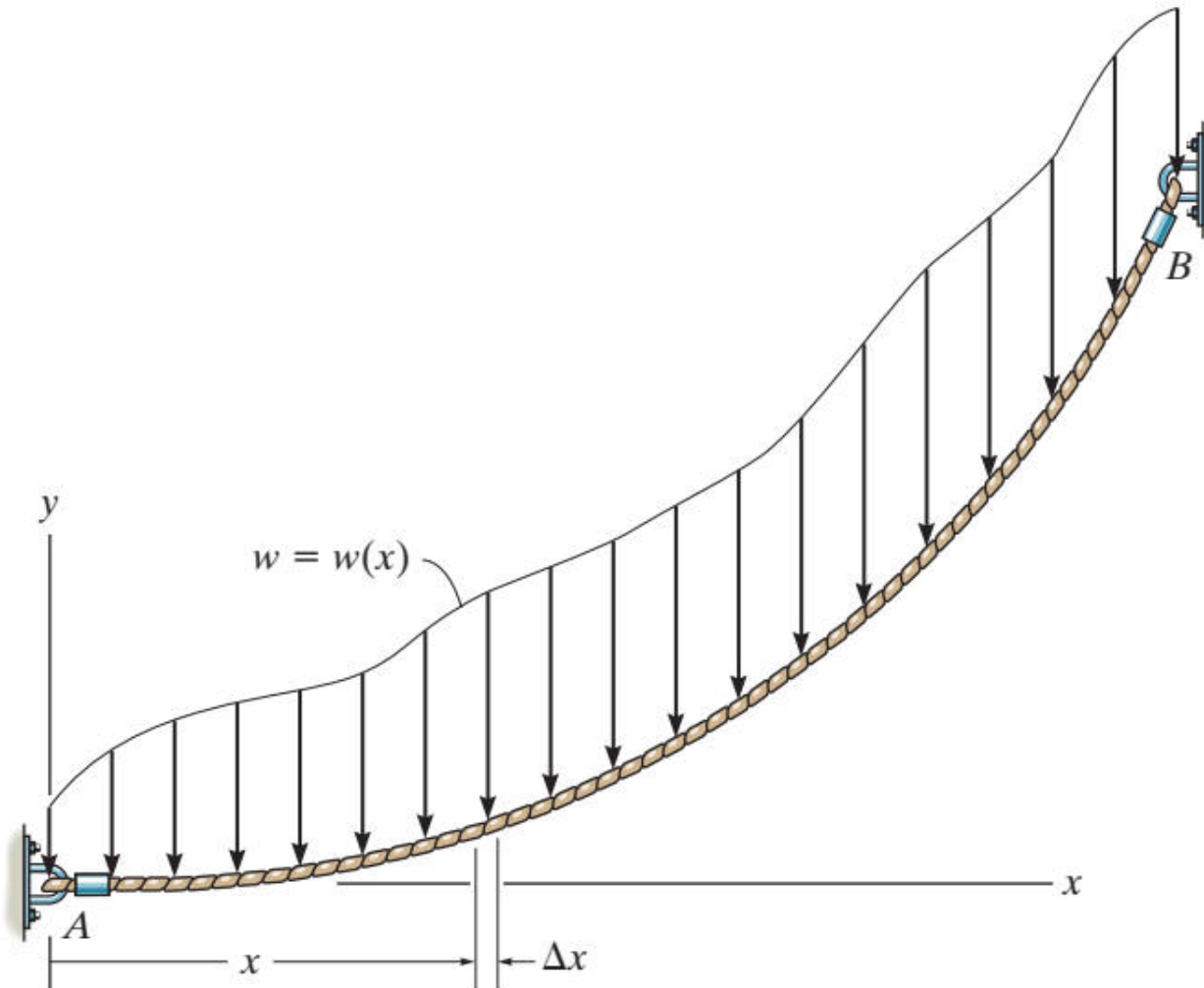


Cables Subjected to Distributed Loads



The cable and suspenders are used to support the uniform load of a gas pipe which crosses the river.

Cables Subjected to Distributed Loads



Cables Subjected to Distributed Loads

$$\sum F_x = 0 \quad -T \cos \theta + (T + \Delta T) \cos(\theta + \Delta \theta) = 0$$

$$-T \cancel{\cos \theta} + T \cancel{\cos \theta} \cos \Delta \theta - T \sin \theta \sin \Delta \theta + \Delta T \cos \theta \cos \Delta \theta$$

$$- \Delta T \sin \theta \sin \Delta \theta = 0$$

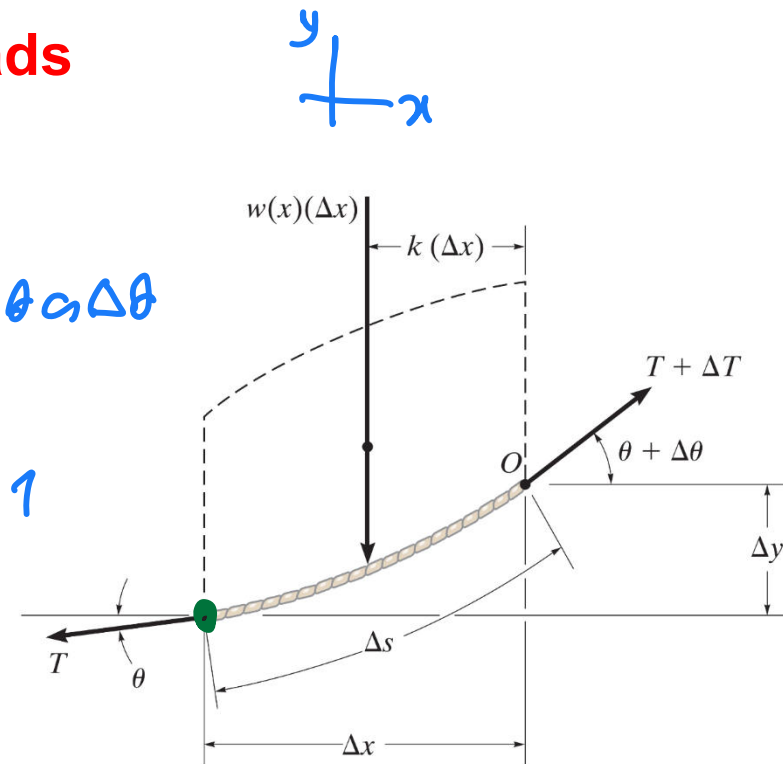
$$\Delta x \rightarrow 0, \quad \Delta \theta \rightarrow 0 \rightarrow \sin \Delta \theta = \Delta \theta \rightarrow \cos \Delta \theta = 1$$

$$-T \sin \theta \Delta \theta + \Delta T \cos \theta = 0$$

$$\rightarrow \boxed{dT \cos \theta = 0}$$

$$\sum F_y = 0 \quad (T + \Delta T) \sin(\theta + \Delta \theta) - T \sin \theta = w(x) \Delta x \rightarrow$$

$$\sin \theta + 0$$



$$\Delta T \sin \theta = w(x) \Delta x$$

Cables Subjected to Distributed Loads

Governing equations:

$$dT \cos \theta = 0 :$$

$$\frac{d(T \cos \theta)}{dx} = 0$$

$$dT \sin \theta = W(x) dx :$$

$$\frac{d(T \sin \theta)}{dx} - w(x) = 0$$

$$\text{from Geometry :}$$

$$\frac{dy}{dx} = \tan \theta$$

Cables Subjected to Distributed Loads

Then, horizontal component of tensile force at *any point* along the cable:

$$T \cos \theta = \text{constant} = F_H$$

Moreover,

$$T \sin \theta = \int w(x) dx$$

$$\tan \theta = \frac{dy}{dx} = \frac{1}{F_H} \int w(x) dx$$

$$y = \frac{1}{F_H} \int \left(\int w(x) dx \right) dx$$

Example: The cable of a suspension bridge supports half of the uniform road surface between the two towers at A and B . If this distributed loading is w_0 , determine the maximum force developed in the cable and the cable's required length. The span length L and sag h are known.

$$y = \frac{1}{F_H} \iint w_0 dx dx = \frac{1}{F_H} \left(\frac{1}{2} w_0 x^2 + C_1 x + C_2 \right)$$

$$y(x=0) = 0 \rightarrow C_2 = 0$$

$$dy/dx(x=0) = 0 \rightarrow \left. \frac{1}{F_H} (w_0 x + C_1) \right|_0 = 0 \rightarrow C_1 = 0$$

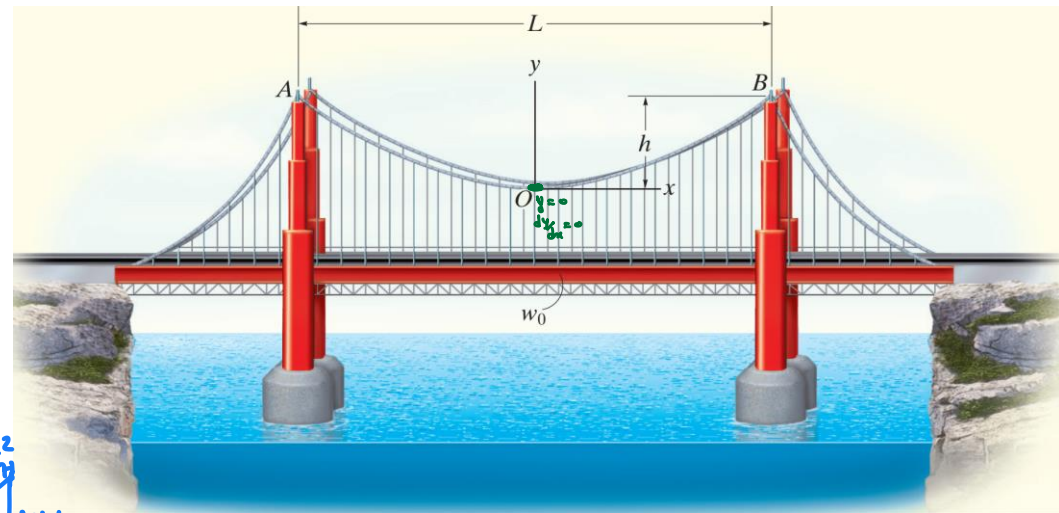
$$y(x) = \frac{w_0 x^2}{2F_H} ; T_{max} = ? \quad F_H = T \cos \theta$$

$$T_{max} = \frac{F_H}{(\cos \theta)_{min}} = \frac{F_H}{\cos \theta_{max}} ; y\left(\frac{L}{2}\right) = \frac{w_0 L^2}{8F_H}$$

$$\frac{dy}{dx} = \tan \theta \quad \frac{dy}{dx} = \frac{w_0 x}{F_H} \rightarrow \tan \theta_{max} = \frac{w_0 L}{2F_H}$$

$$T_{max} = \frac{F_H}{\cos \theta_{max}} = \frac{F_H}{\frac{2F_H}{\sqrt{w_0^2 L^2 + 4F_H^2}}} = \frac{1}{2} \sqrt{w_0^2 L^2 + 4F_H^2}$$

$$y\left(\frac{L}{2}\right) = \frac{w_0 L^2}{8F_H} = h \rightarrow F_H = \frac{w_0 L^2}{8h}$$



Example: The cable of a suspension bridge supports half of the uniform road surface between the two towers at A and B . If this distributed loading is w_0 , determine the maximum force developed in the cable and the cable's required length. The span length L and sag h are known.

$$L_{OB} = \int_0^B ds, \quad \frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}, \quad ds = \sqrt{dx^2 + dy^2}$$

$$ds = dx \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \quad \frac{dy}{dx} = \frac{w_0 x}{F_H}$$

$$L_{OB} = \int_0^{L/2} \sqrt{1 + \left(\frac{w_0 x}{F_H}\right)^2} dx \quad x = \frac{F_H}{w_0} \sinh u$$

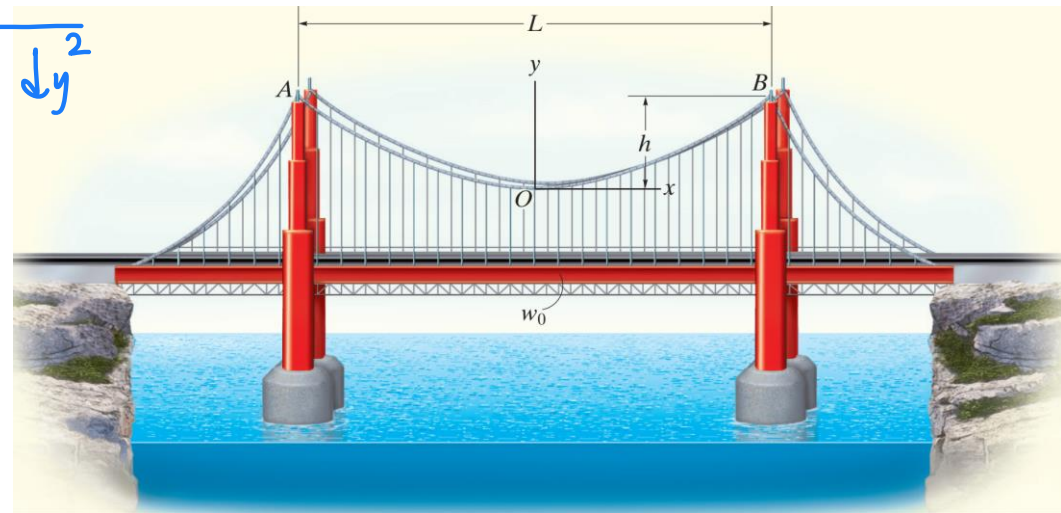
$$dx = \frac{F_H}{w_0} (\cosh u) du \quad \cosh^2 u - \sinh^2 u = 1$$

$$\int (\cosh u) \cdot \frac{F_H}{w_0} \cosh u du = \int \frac{F_H}{w_0} \cosh^2 u du$$

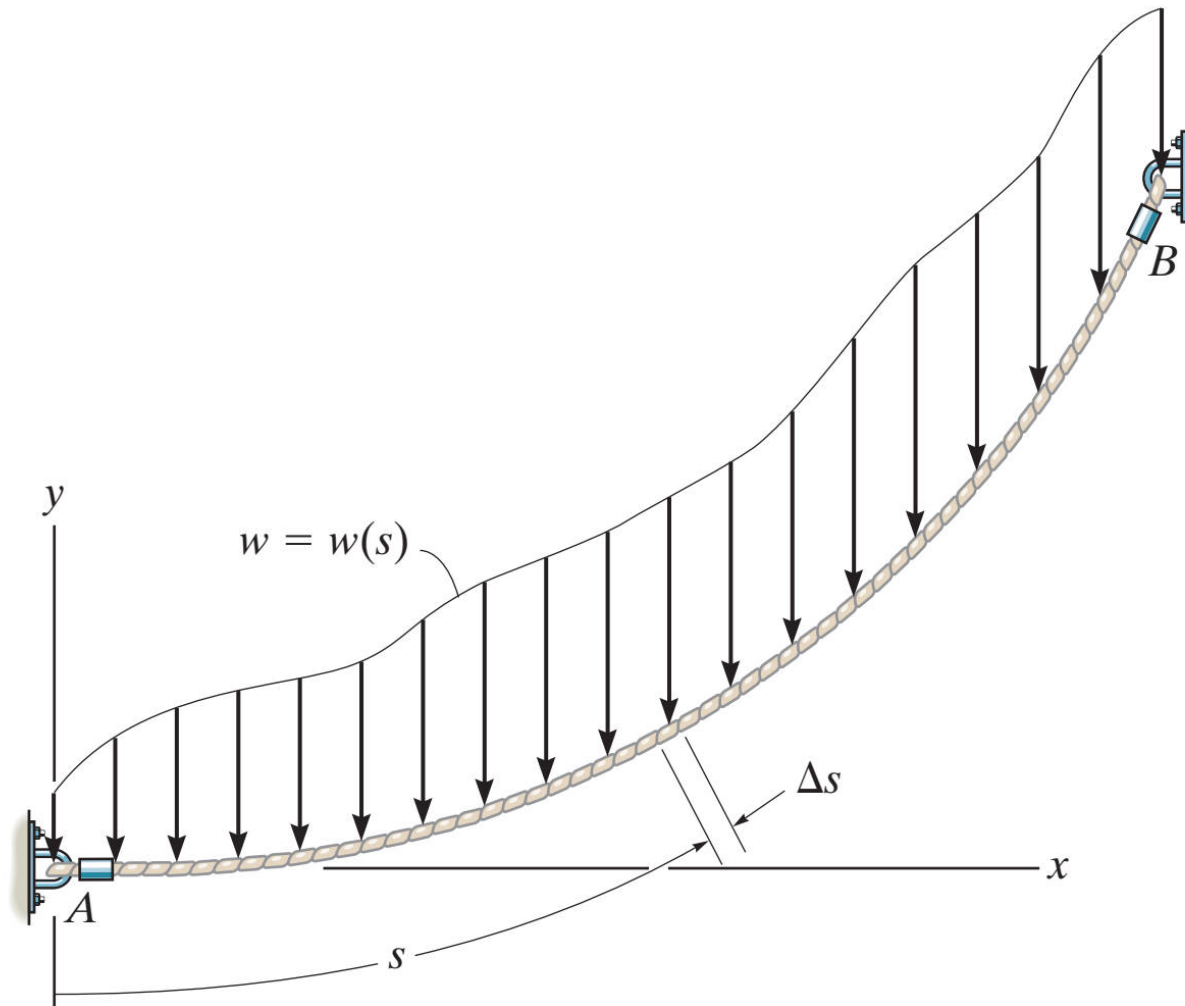
$$\cosh u = \frac{e^u + e^{-u}}{2} \rightarrow \int \frac{F_H}{w_0} \left(\frac{e^{2u} + e^{-2u} + 2}{4} \right) du = \frac{F_H}{4w_0} \left(\frac{1}{2} e^{2u} - \frac{1}{2} e^{-2u} + 2u \right)$$

$$\rightarrow \frac{F_H}{4w_0} \left[\sinh 2u + 2u \right] \quad u = \sinh^{-1} \left(\frac{w_0 x}{F_H} \right)$$

$$\frac{2w_0 x}{F_H}$$



Cables Subjected to their Own Weight



Cables Subjected to their Own Weight

$$T \cos \theta = F_H$$

$$T \sin \theta = \int w(s) ds$$

$$\frac{dy}{dx} = \frac{1}{F_H} \int w(s) ds$$

$$ds = \sqrt{dx^2 + dy^2}$$

Cables Subjected to their Own Weight

Finally, the relation between x and s parameters can be obtained as:

$$x = \int \frac{ds}{\left[1 + \frac{1}{F_H^2} \left(\int w(s) ds \right)^2 \right]^{1/2}}$$

Cables Subjected to their Own Weight

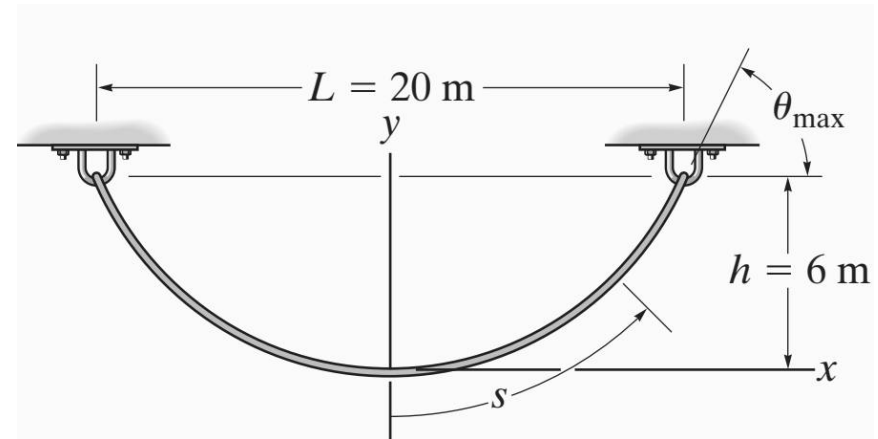


Electrical transmission towers must be designed to support the weight of the suspended power lines.

Example: Determine the shape of the curve, the length, and the maximum tension in the uniform cable shown below. The cable has a weight per unit length of $w_0 = 5 \text{ N/m}$.

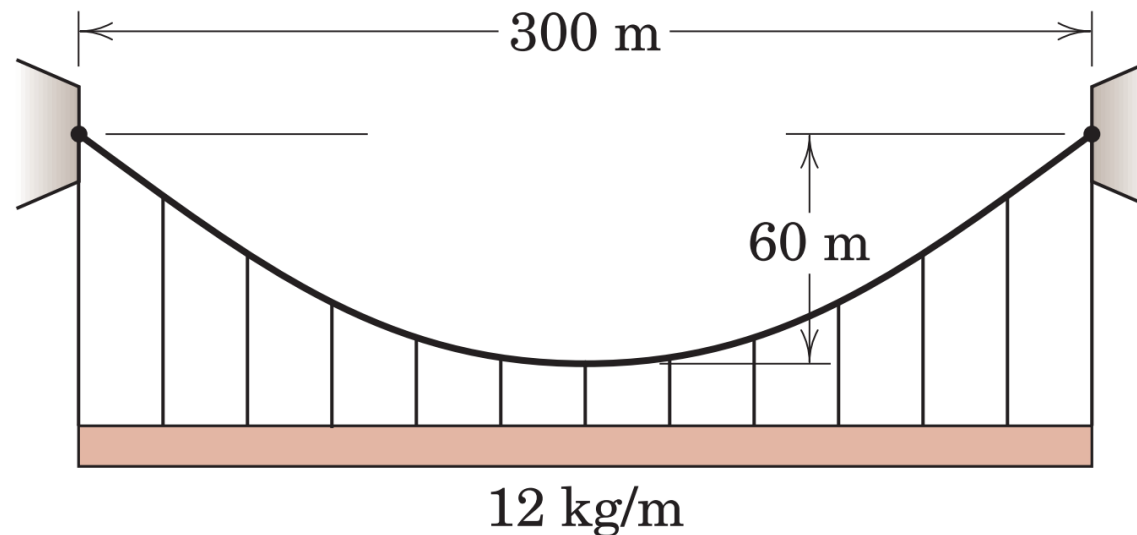
Hints:

$$\operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1}), \quad \frac{d}{dx} \operatorname{arsinh} x = \frac{1}{\sqrt{x^2 + 1}}$$



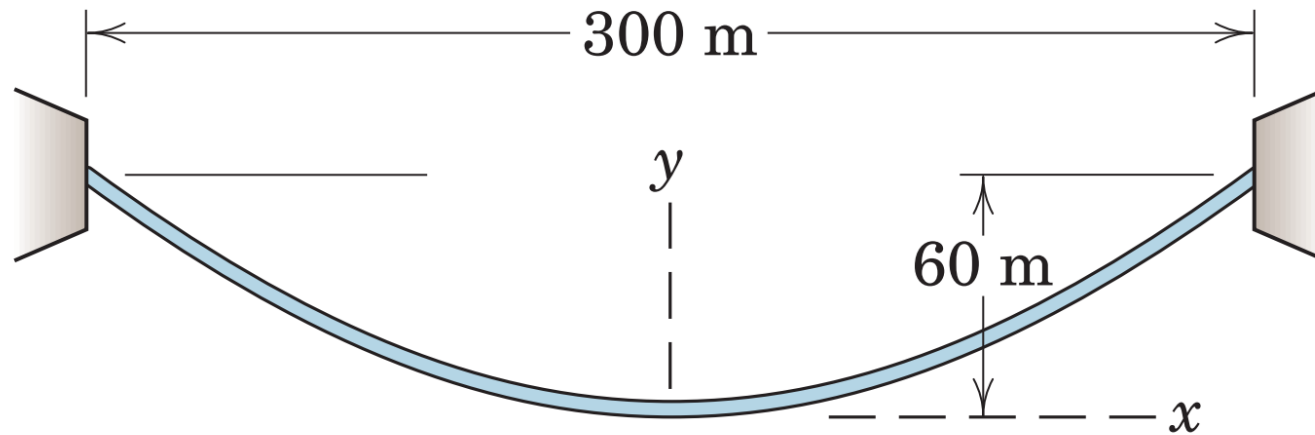
Practice at Home

1. The light cable supports a mass of 12 kg per meter of horizontal length and is suspended between the two points on the same level 300 m apart. If the sag is 60 m, find the tension at mid-length, the maximum tension, and the total length of the cable.



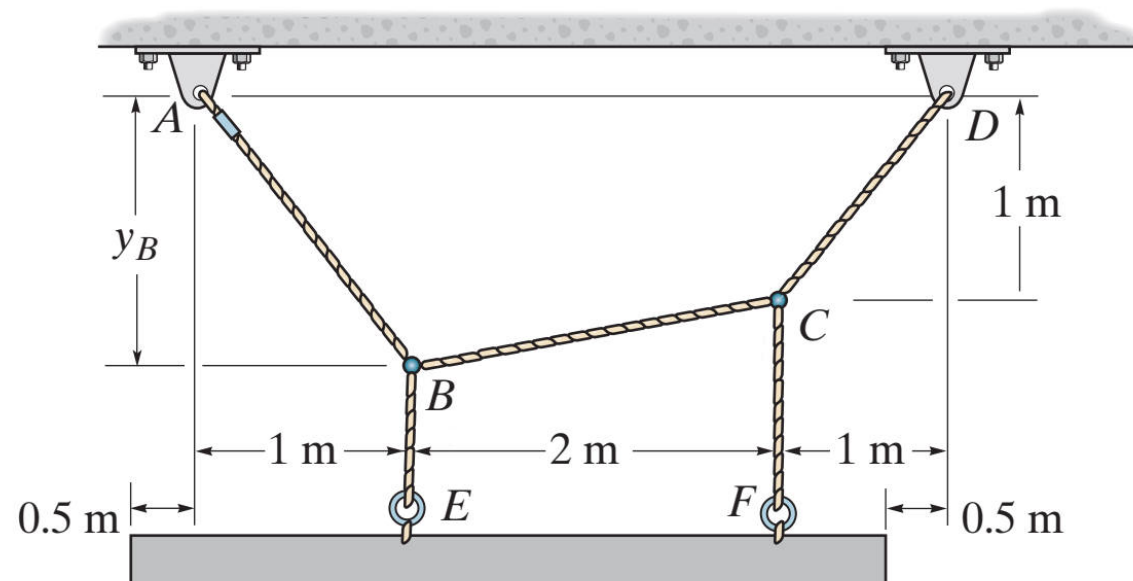
Practice at Home

2. As shown, a cable which has a mass of 12 kg per meter of its own length and supports its own weight only. The cable is suspended between two points on the same level 300 m apart and has a sag of 60 m. Find the tension at mid-length, the maximum tension, and the total length of the cable.



Practice at Home

3. Cable $ABCD$ supports the 120-kg uniform beam. Determine the maximum tension in this cable and the sag of point B .



Practice at Home

4. A cable weighing 25 newtons per meter of length is suspended from point A and passes over the small pulley at B . Calculate the mass m of the attached cylinder which will produce the sag of 9 m. Also determine the horizontal distance from A to C . Because of the small sag-to-span ratio, use the approximation of a parabolic cable.

